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1. Introduction

Optical coherence tomography angiography (OCTA) is an extension of OCT that visualizes vascular perfusion using the motion contrast from red blood cells (RBCs). Over the years, OCTA has demonstrated widespread acceptance and use for clinical evaluations of vascular features in relation to disease [1]. OCT angiograms can be used to evaluate characteristics on the scale of vascular layers by measuring vascular density or non-perfusion zones, or it can be used to examine individual vessels with measurements of diameter or tortuosity. Nevertheless, OCT is prone to certain artifacts which specifically affect visualization of the vasculature. These artifacts may lead to underestimations of measurements of vessel diameter and density. When viewed in cross-section, large vessels often demonstrate an hourglass appearance with stronger scattering in the center than at the sides (Figure 1A). This appearance is due to orientation effects of RBCs, which align with the blood flow, causing a reduced scattering cross-section at the sides of the vessels. To compensate for these artifacts, OCT contrast agents such as Intralipid have previously been used to increase intravascular intensity signals, particularly in regions where RBC scattering is diminished. Here we examine different methods of measuring vessel size from in vivo OCT scans of mouse retinal vasculature, and we use a contrast agent to obtain ground truth vessel diameter measurements for comparison. We finally propose ways of mitigating or correcting for underestimations of vessel diameter related to typical OCT artifacts.

Optical coherence tomography (OCT) is a non-invasive optical imaging method, which uses interferometry to measure the transit time of light that is backscattered from a sample. OCT has found widespread clinical use in ophthalmology with XXX number of systems and XXX OCT scans performed annually around the world. OCT angiography (OCTA) is a functional extension of OCT which highlights motion between repeated OCT scans to generate images of vascular networks. Typically, OCTA signals are first calculated from the OCT intensity and/or phase information, segmented in 3D manually, algorithmically, or using a deep learning approach, and finally compressed into a 2D image using a maximum intensity projection. These 2D vascular maps can then be binarised by using a global or adaptive threshold, and finally vascular metrics such as vessel density and diameter can be computed. Of these processing stages, the thresholding step is perhaps the most important. There are dozens of different thresholding methods that have been proposed over the years and each of these may produce different results in the binary map, which would in turn influence the quantification of vascular metrics. In this work, we examine an alternate approach for extracting quantitative vessel diameter by fitting a model to the OCTA data before binarisation to avoid thresholding-based effects.

In addition to thresholding-based variability, we hypothesize that OCTA metrics are also affected by OCT artifacts caused by the non-uniform backscattering profiles of red blood cells (RBCs). Because RBCs have a distinct bi-concave disc shape, they more effectively backscatter light when aligned perpendicular to the beam, rather than parallel to it. It has already been observed that in large vessels, a so-called hourglass artifact appears when viewed cross-sectionally. This is caused by the alignment of the RBCs with the vessel walls while under laminar flow, where the RBCs are perpendicular to the beam in the center of the vessels and parallel to the beam at the edges of the vessels. This artifact artificially suppresses the OCT and OCTA signals at the edges of the vessels, making them appear more narrow when projected into an en-face view, which is the preferred way of displaying angiographic data. Here we use contrast enhanced OCT (CE-OCT) to investigate how significant these effects are and how they can be corrected.

2. Methods and Materials

2.1. Imaging system

All imaging was performed using a custom-built polarisation-sensitive spectral-domain OCT ophthalmoscope, described in detail previously [2]. The system was centred at 840 nm with a spectral bandwidth of 100 nm at full width half maximum (FWHM), providing an axial resolution of approximately 5.1 µm in air and 3.8 µm in tissue (assuming a refractive index of 1.35). A sensitivity of 96 dB was achieved at an A-scan rate of 80 kHz and a beam diameter of 0.5 mm at the pupil plane (power at cornea = 2.85 mW) [3]. This system was employed to acquire volumetric OCT data from mouse retinas before, during, and after an intravascular contrast agent administration, as well as before and after functional stimulation of the mouse retina using a custom 502 nm light-emitting diode (LED) ring flashing at 10 Hz (power at cornea = 26.6 μW). This wavelength was selected to target the peak absorption of mouse rhodopsin (~500 nm), enabling predominantly rod-driven photoreceptor stimulation [4].

2.2. Experimental protocol

Three mouse models were used in this study: VLDLR knockout mice (B6;129S7-*Vldlr^{tm1Her}*/J, The Jackson Laboratory, Bar Harbor, USA), C57BL/6 mice (C57BL/6, The Jackson Laboratory, Bar Harbor, USA), and SOD1 transgenic mice (strain details, The Jackson Laboratory, Bar Harbor, USA), with n = XX mice per group. All imaging sessions were conducted under general anaesthesia using either isoflurane or a ketamine/xylazine combination. Isoflurane was induced at 4% in oxygen for 4 minutes and subsequently maintained at 2% in oxygen at a flow rate of 1.5–2 L/min. In cases where isoflurane alone did not provide adequate sedation, an intraperitoneal injection of ketamine/xylazine cocktail (100 mg/kg ketamine, 6 mg/kg xylazine; 10 mL/kg body weight) was administered. To maintain core body temperature of the mice throughout imaging, they were positioned on a heating pad. Mouse pupils were dilated with one drop of tropicamide (5 mg/mL, Agepha Pharma s.r.o., Senec, Slovakia) per eye, and artificial tear drops (Oculotect, ALCON Pharma GmbH, Freiburg im Breisgau, Germany) were applied at regular intervals to prevent the cornea from drying out.

Intralipid has been shown to be a highly scattering contrast agent that fills the vascular lumen uniformly, thereby compensating for the artefact caused by the diminished backscattering at vessel peripheries caused by RBC orientation effects [3, 5, 6]. For this study, Intralipid 20% (3 mL/kg body weight) was administered as a bolus injection via the tail vein. Volumetric angiography data were acquired over a 1 mm × 1 mm field of view centred on the optic nerve head (ONH), with a scan density of 500 A-scans per B-scan and five repeated B-scans at each of 400 slow-axis positions (acquisition time approximately 16 seconds). For this study, pre- and post-injection datasets and pre- and post-stimulation datasets were acquired. These volumetric data were used solely for qualitative visualisation of the contrast enhancement effect across the retinal vascular plexuses and were not used for quantitative diameter analysis. Additionally, a dynamic-contrast OCT (DyC-OCT) protocol [7] was employed during the contrast injection or during stimulation, consisting of 2000 repeated B-scans over approximately 16 seconds (interscan time 8.2 ms) at the same cross-sectional location over a 1 mm lateral field of view. DyC-OCT scans allow us to track the same vessels over time to precisely measure the effects of the contrast agent or the stimulation with a high degree of spatial coregistration. All quantitative vessel diameter measurements reported in this study were derived from the DyC-OCT data.

Functional flicker stimulation was performed in a C57BL/6 mouse to assess whether the choice of diameter measurement method affects the detection of physiological vascular responses. For this experiment, ketamine/xylazine cocktail was chosen over isoflurane to minimise anaesthesia-induced vasodilation. A DyC-OCT acquisition was performed during which the retina was unstimulated for approximately 4 seconds (baseline), followed by 10 Hz flicker stimulation from the LED ring for approximately 7.5 seconds, and a post-stimulation period of approximately 4.5 seconds (I wrote this to match the pre- and post-stim lines in the stimulation result plots).

All animal procedures were approved by the ethics committee of the Medical University of Vienna and the Austrian Federal Ministry of Education, Science, and Research (BMBWF/66.009/0272-V/3b/2019 and GZ 2024-0.802.524).

2.3. Post-processing and analysis

Cross-polarised OCT intensity data from the retinal pigment epithelium (RPE) were used for inter-frame alignment and B-scan flattening [3, 8]. Angiograms were derived by applying the complex subtraction OCT angiography (OCTA) algorithm [9, 10]. We evaluated multiple combinations of methods for quantifying vessel diameter from the OCTA data to determine which combinations of steps work best under different conditions. Specifically, we looked at two types of data projections (maximum intensity and mean intensity) to generate vessel profiles and several different quantifications of those profiles including a traditional binarisation, a Gaussian model approach, and a Generalised-Gaussian (GG) model approach. These methods represent commonly used methods in the literature for OCTA quantification [11–14]. The different methods were evaluated against the ground truth as described in this section. All data processing and analysis was performed in MATLAB (R2019b, MathWorks, Natick, MA, USA).

2.3.1. Ground truth calculation

Ground truth vessel diameters were obtained from contrast-enhanced DyC-OCT B-scan angiograms, where the uniformly scattering Intralipid fills the full vascular lumen regardless of RBC orientation. The width and the height of the vessel were manually measured from the vessel's cross-section. The ground truth diameter was defined as the minimum of the two measurements to account for off-axis cross-sectioning of the vessel, which produces a more elliptical appearance in the B-scan. For cases where the vessel was perpendicular to the B-scan, the height and width were in agreement. Because the lateral diameter is studied here, we defined a correction factor of $height/width$ for cases where the width was larger than the height due to skew effects and was equal to 1 otherwise. By multiplying this correction factor against diameter measured from the ROI projection, we ensured that the measured vessel diameter reflected the true size of the vessel.

2.3.2. Vessel projection calculation

For each vessel of interest, a rectangular region of interest (ROI) encompassing the vessel cross-section, along the lateral (x) and axial (z) dimensions, was selected from the DyC-OCT B-scan angiograms. En face intensity profiles were then generated by projecting the A-scans within this ROI along the axial (z) direction yielding a time-stack of 1-D lateral intensity profiles for each vessel. Two projection methods were compared: maximum intensity projection (MIP), which selects the highest intensity value along the depth axis of each ROI, and mean intensity projection (MeIP), which averages the values along each ROI. Each projection resulted in a time-stack of one-dimensional intensity profiles for each vessel, which were then used for subsequent diameter quantification. For the model-based approaches, the profiles were normalised to a range of 0–1 to ensure a good fit.

2.3.3. Model-based vessel diameter calculation

The two parametric models were fitted (Fig. 1a) to the one-dimensional intensity profiles from the previous step, using nonlinear least-squares regression in MATLAB.

Standard Gaussian function:

$$f(x) = A \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) + C \tag{1}$$

where A is the peak amplitude, μ is the centre position, σ is the standard deviation, and C is a constant baseline offset.

Generalised-Gaussian (GG) function:

$$f(x) = A \exp\left(-\left(\frac{|x-\mu|}{\alpha}\right)^{\beta}\right) + C \tag{2}$$

where A is the peak amplitude, μ is the centre position, α is the scale parameter, β is the shape parameter, and C is a constant baseline offset. Given the imaging system configuration, all vessels are expected to produce at minimum a Gaussian intensity profile in the B-scan cross-section, and so profiles sharper than Gaussian profile ($\beta < 2$) are therefore not technically meaningful. The shape parameter β was hence constrained to $\beta \geq 2$ during fitting. With this constraint, the GG accommodated to profiles ranging from Gaussian ($\beta = 2$, where $\sigma = \alpha/\sqrt{2}$) to increasingly flat-topped ($\beta > 2$).

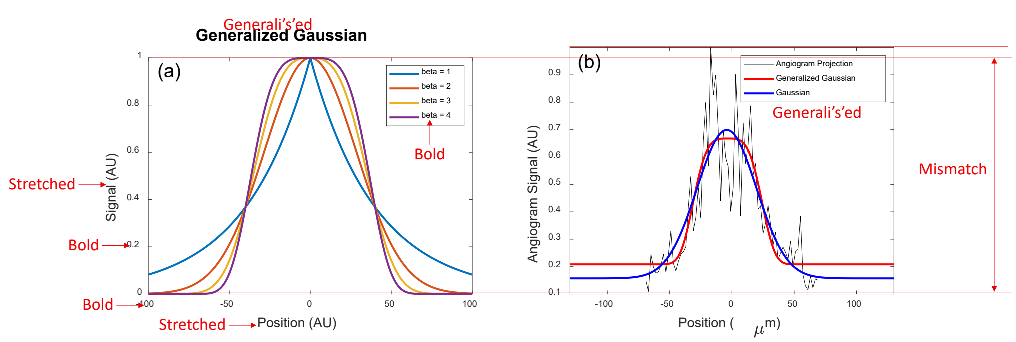

Fig. 1. (a) GG function for shape parameters $\beta = 1, 2, 3$, and 4. At $\beta = 2$ the function reduces to a standard Gaussian; higher β values produce increasingly flat-topped profiles. (b) Example en face vessel intensity profile (black) with standard Gaussian (blue) and GG (red) fits.

Two standard width metrics were calculated from the fitted models: the full width at half maximum (FWHM) and the $1/e^2$ width. For the standard Gaussian fit, the $1/e^2$ width was calculated as:

$$w_{1/e^2,\,\mathrm{G}} = 4\sigma \tag{3}$$

The FWHM for a Gaussian is related to the $1/e^2$ width by a fixed proportionality:

$$\mathrm{FWHM}_{\mathrm{G}} = 2\sigma\sqrt{2\ln 2} = \frac{\sqrt{2\ln 2}}{2}\, w_{1/e^2,\,\mathrm{G}} \tag{4}$$

and therefore only the $1/e^2$ width was calculated directly from the fit, with the FWHM obtainable via this known relationship.

For the GG fit, no such linear relationship exists between the FWHM and $1/e^2$ width, since both depend on the shape parameter β. Both metrics were therefore calculated independently as:

$$\mathrm{FWHM}_{\mathrm{GG}} = 2\alpha(\ln 2)^{1/\beta} \tag{5}$$

$$w_{1/e^2,\,\mathrm{GG}} = 2\alpha \cdot 2^{1/\beta} \tag{6}$$

2.3.4. Binary mask based vessel diameter calculation

For comparison with OCTA processing pipelines that rely on image binarisation, vessel diameters were also measured from binary masks generated by global thresholding. A 3D averaging filter with a kernel size of $50 \times 50 \times 10$ voxels (lateral x × axial z × temporal t) was applied to the DyC-OCT angiogram volumes to reduce noise. To get a binarised black and white (BW) mask, the maximum angiogram intensity within these smoothed volumes were calculated and five different threshold values were defined as fixed percentages of these maximum: BW-1 = XX%, BW-2 = XX%, BW-3 = XX%, BW-4 = XX%, and BW-5 = XX%. Each threshold was applied to the unsmoothed en face projection (MIP and MeIP) to generate a binary mask, and the vessel diameter was measured as the number of pixels above the threshold across the lateral vessel cross-section.

2.3.5. Statistical analysis

To evaluate the accuracy and precision of all diameter measurement approaches, the calculated diameter values were plotted against the ground truth for each vessel across three conditions: pre-contrast, peak contrast, and post-contrast. A linear regression was fitted to each condition, yielding the slope (m) and coefficient of determination (R^2). A slope of $m = 1$ indicated perfect agreement with the ground truth, and R^2 quantified the goodness of fit of the linear model. Additionally, the coefficient of variation (CoV) was computed for each method to assess measurement precision (repeatability). For the model-based methods, six combinations were evaluated: two projection methods (MIP and MeIP) by three width metrics ($w_{1/e^2,\,\mathrm{G}}$, $\mathrm{FWHM}_{\mathrm{GG}}$, and $w_{1/e^2,\,\mathrm{GG}}$). For the binarisation-based methods, diameter was measured at five threshold levels (BW-1 through BW-5) for both MIP and MeIP. For the functional flicker stimulation experiment, the Pearson correlation coefficient between measured diameter values and angiogram intensity was computed for each method and vessel, to assess whether the diameter measurements were affected by changes in angiogram intensity. Measurements with $p < 0.05$ were considered statistically significant. All statistical analyses were performed in MATLAB (R2019b, MathWorks, Natick, MA, USA).

3. Results

3.1. Qualitative assessment of vessel underestimation

Figure 2 shows OCTA MIP en face of the superficial vascular plexus (SVP), intermediate capillary plexus (ICP), and deep capillary plexus (DCP) before and after contrast injection. Across all

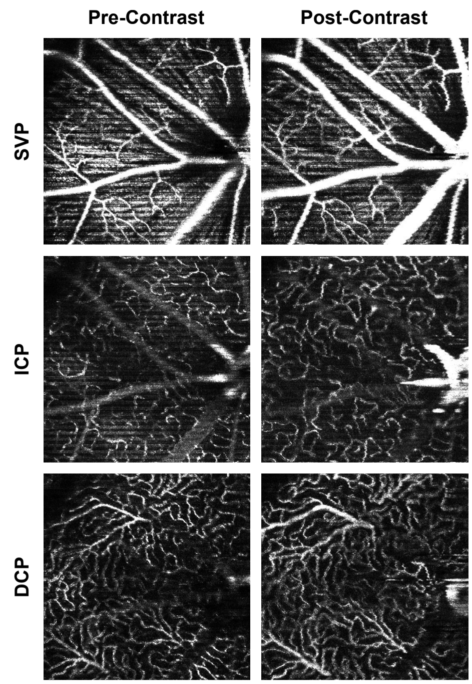

Fig. 2. OCTA MIP en face of the SVP, ICP, and DCP before (left) and after (right) the administration of Intralipid 20%. Vessels appear visibly wider and brighter post-contrast across all layers, most prominently in the larger vessels.

three layers, vessels appeared visibly wider and brighter post-contrast injection, consistent with previous literature showing that OCTA underestimates vessel diameter in the absence of a contrast agent [3]. This effect was most pronounced in the larger vessels, especially in the SVP, where the major retinal arteries and veins appeared noticeably thicker after contrast administration. In the ICP and DCP, the capillary networks also showed improved visibility, though the effect was less dramatic.

DyC OCTA projections helped visualise the temporal changes in vessel diameter pre- and post-contrast injection (Fig. 3). Pre-contrast injection, the vessel cross-sections appeared narrower than their true diameter (Fig. 3a), which could be due to lateral edge signal suppression by the RBC orientation artefact [15]. At peak contrast, the Intralipid filled the vascular lumen uniformly and showed the full extent of the vessel cross-section (Fig. 3b). However, the high scattering nature of the contrast agent produced light diffusion beyond the vessel wall, resulting in an apparent vessel diameter that slightly exceeded the true anatomical diameter. In the post-contrast phase, the signal stabilised as the initial bolus cleared, and the vessel boundaries remained more visible than before injection without the overestimation observed at peak contrast (Fig. 3c). The MIP and the corresponding binary mask over time (Figs. 3d–e) showed the observed temporal changes in vessel diameter across the three phases. The mean OCTA intensity (Fig. 3f) showed a sharp increase post-contrast injection, followed by a decrease to a new steady state that remained elevated above the pre-contrast baseline. These observations confirm that the contrast agent does significantly enhance vessel visibility and diameter, and that the post-contrast images show improved vessel delineation compared to pre-contrast.

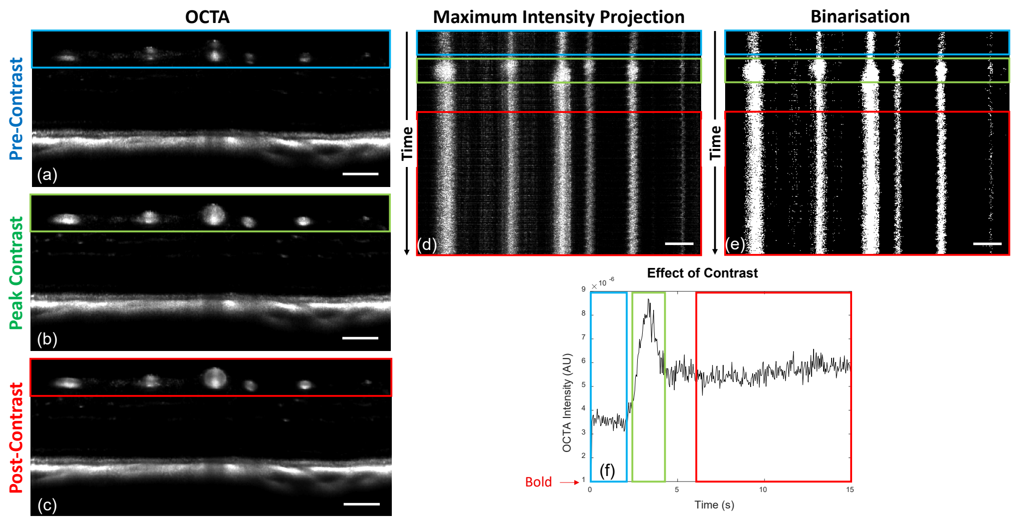

Fig. 3. DyC-OCT B-scan angiograms at (a) pre-contrast, (b) peak contrast, and (c) post-contrast timepoints. (d) MIP and (e) binarisation of the vessel cross-sections over time, with coloured boxes indicating the pre-contrast (blue), peak-contrast (green), and post-contrast (red) windows. (f) Mean OCTA intensity over time, showing the temporal intensity dynamics. All scale bars are 100 µm.

3.2. Accuracy and precision of diameter measurements

The accuracy (m, R^2) and precision (CoV) of all diameter measurement methods are summarised in Table 1.

All model-based methods underestimated vessel diameter in pre-contrast data, with slopes ranging from 0.42 to 0.73 (Table 1). Contrast enhancement improved accuracy across all metrics, though the degree of improvement varied. The FWHM_{GG} metric remained well below unity even post-contrast. The $w_{1/e^2,\text{G}}$ with MIP slightly overestimated post-contrast ($m = 1.12$–1.15), while MeIP yielded values closer to unity ($m = 1.04$). The $w_{1/e^2,\text{GG}}$ with MeIP provided the best overall agreement, achieving a slope closest to unity with $m = 1.001$ ($R^2 = 0.97$) post-contrast, which was the closest to the identity line of all tested model-based methods. Across all metrics, MeIP mostly outperformed MIP in both accuracy and linearity (Table 1, Fig. 4).

For binary mask based methods, the slope decreased with increasing threshold across all contrast conditions and projections (Table 1). This degradation was steeper for MIP than for MeIP. MIP slopes dropped from near unity at BW-1 to well below it at BW-5, whereas MeIP slopes declined more gradually and remained closer to unity across the threshold range. MeIP slopes were consistently higher than MIP at every threshold and contrast condition, though at lower thresholds and at peak and post-contrast, MeIP tended to slightly overestimate vessel diameter. The R^2 values followed a similar pattern. For MIP, pre- and post-contrast R^2 decreased substantially with increasing threshold, whereas for MeIP, R^2 remained relatively stable. Overall, MeIP provided more reliable binarisation-based diameter measurements than MIP.

The precision of each method was assessed via the CoV (Table 1, Fig. 7). For MIP, the binary mask based CoV increased substantially with threshold, whereas the model-based CoV remained stable across all contrast conditions. For MeIP, the binary mask based CoV was notably more stable across thresholds than for MIP, and the model-based CoV values were lower overall, though they increased slightly at peak contrast before decreasing post-contrast. Across both methods, MeIP yielded lower CoV values than MIP pre-contrast and more stable values post-contrast. Among the model-based metrics, FWHM_{GG} consistently exhibited the weakest accuracy while

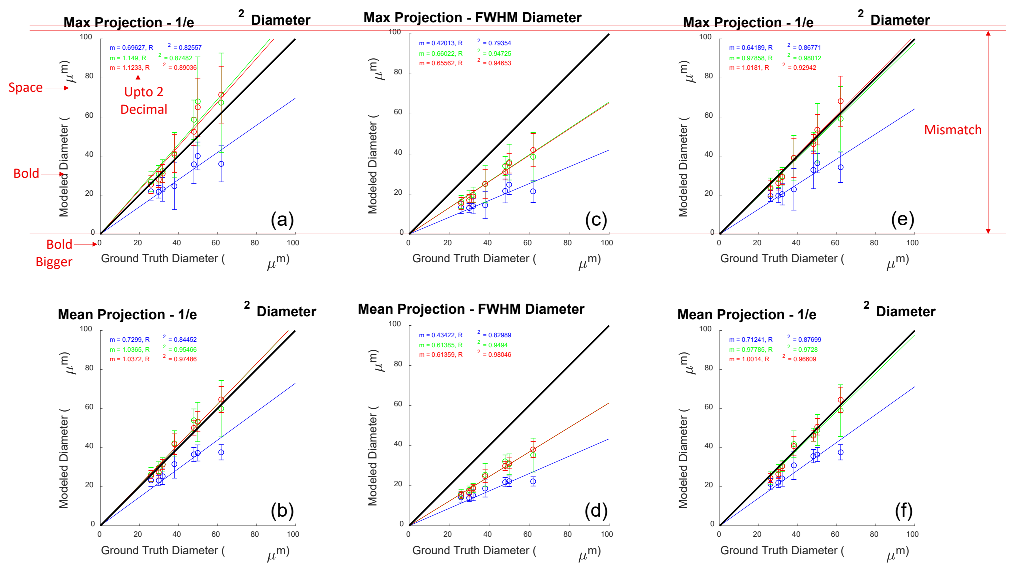

Fig. 4. Modelled versus ground truth vessel diameter using MIP (top row) and MeIP (bottom row). Columns correspond to (a, b) $w_{1/e^2,\mathrm{G}}$, (c, d) $\mathrm{FWHM_{GG}}$, and (e, f) $w_{1/e^2,\mathrm{GG}}$. Linear regression lines and statistics are shown for pre-contrast (blue), peak contrast (green), and post-contrast (red) conditions. The black line indicates the identity line ($m = 1$).

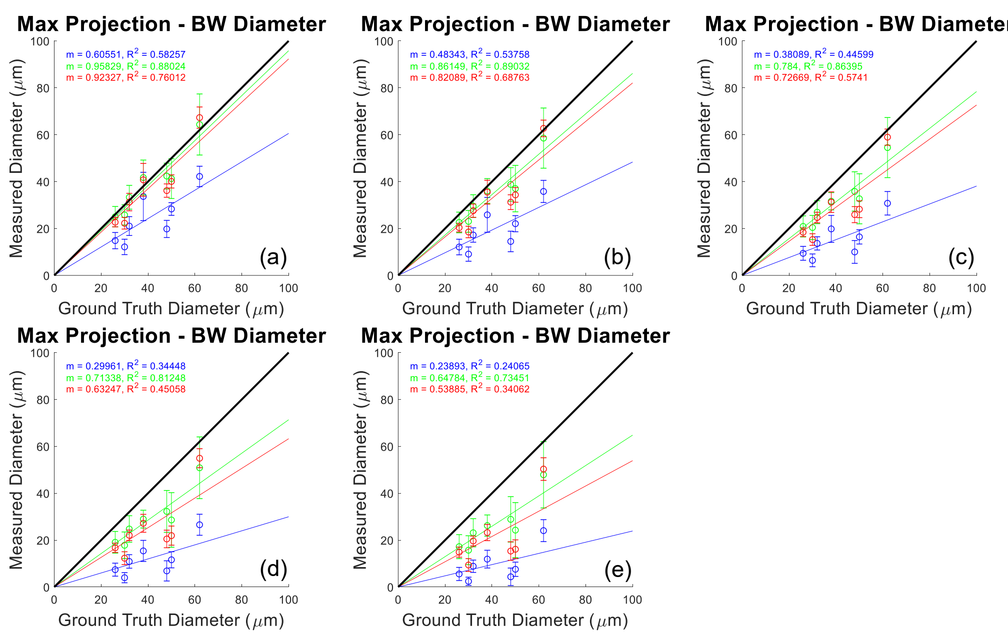

Fig. 5. Vessel diameter measured from binarised masks versus ground truth for MIP at five threshold levels: (a) BW-1 through (e) BW-5. Linear regression lines and statistics are shown for pre-contrast (blue), peak contrast (green), and post-contrast (red) conditions.

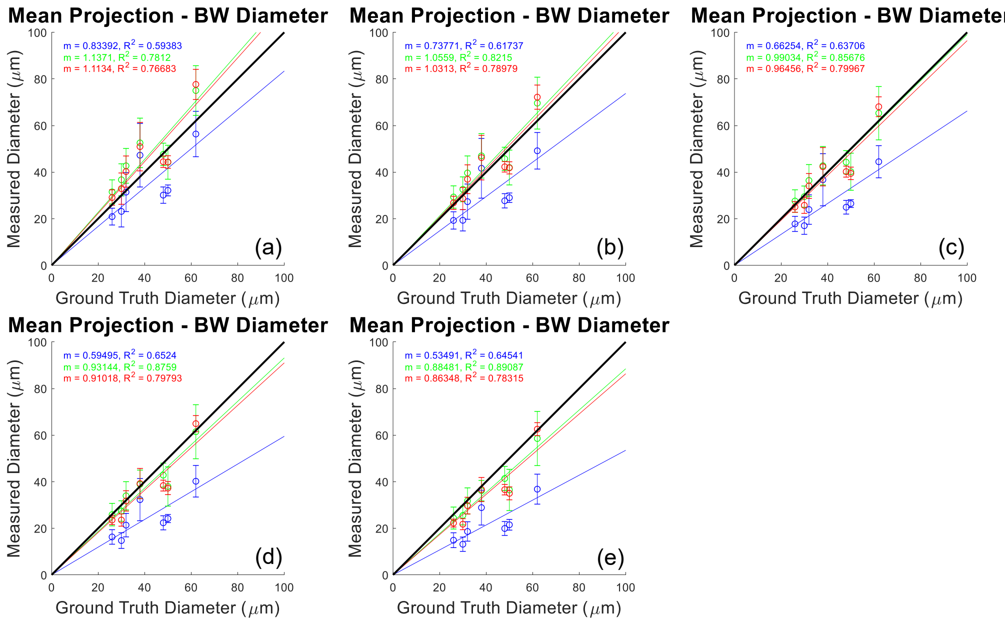

Fig. 6. Vessel diameter measured from binarised masks versus ground truth for MeIP at five threshold levels: (a) BW-1 through (e) BW-5. Linear regression lines and statistics are shown for pre-contrast (blue), peak contrast (green), and post-contrast (red) conditions.

maintaining comparable precision. Overall, model-based methods consistently outperformed binarisation in both accuracy and precision, and MeIP yielded superior performance compared to MIP across all methods.

3.3. Functional flicker stimulation

The model-based diameter time series (Figs. 8a,b) showed increase in diameter post-stimulation for most of the vessels. The binary mask based diameter time series (BW-3;Fig. 8c) showed a mixed trend, where most vessels showed a decrease in diameter post-stimulation. Bar plots of the mean diameter pre- and post-stimulation (Figs. 8d–f) confirmed this pattern. For the Gaussian model, four of five vessels showed significant diameter increases ($p < 0.01$), and the GG model showed significant increases in three of five vessels (Table 2). In contrast, the binary mask based diameter showed significant decreases in four of five vessels.

The angiogram intensity time series (Fig. 8g) showed a visible decline in angiogram intensity during the duration of the DyC-OCT scan across all five vessels. The mean intensity comparison pre- and post-stimulation (Fig. 8h) confirmed that the decrease was significant for all vessels, ranging from −14.1% to −19.9% (Table 2). The binary mask based method showed uniformly positive per-vessel correlations, with an overall correlation of $\rho = 0.523$ ($p < 0.0005$). The Gaussian and GG models yielded overall correlations of $\rho = -0.101$ and $\rho = -0.098$, respectively. These results demonstrated that the model-based diameter calculation provided more accurate and repeatable measurements, and also recovered physiologically meaningful vascular responses that binary mask based methods misrepresented.

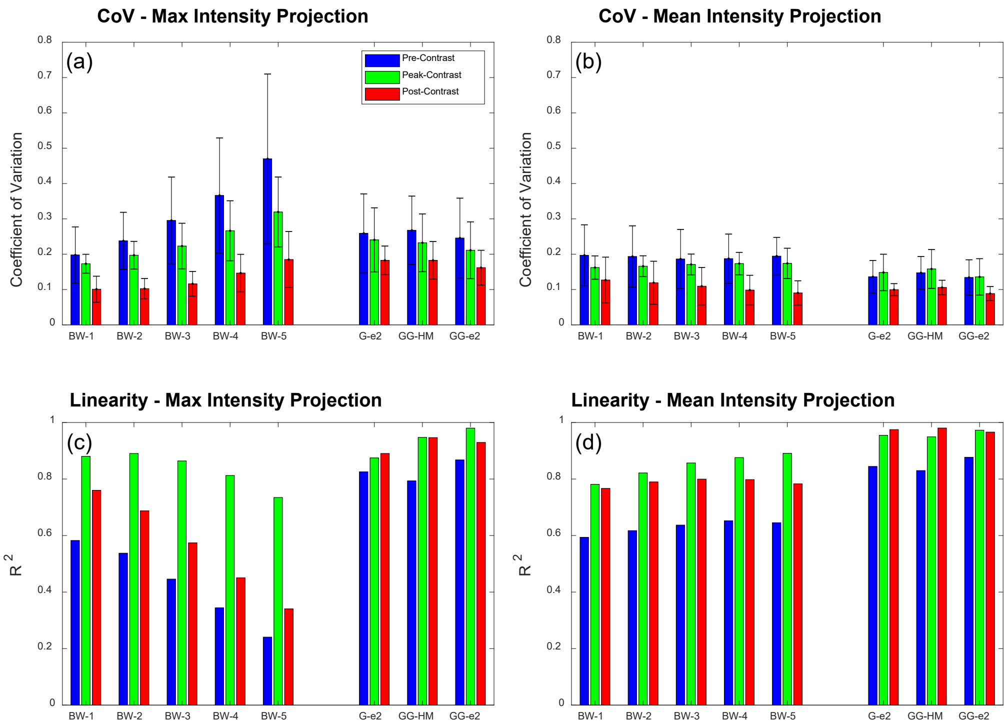

Fig. 7. Summary of accuracy and precision across all diameter measurement methods. (a, b) Coefficient of variation (CoV) and (c, d) coefficient of determination (R^2) for MIP (a, c) and MeIP (b, d), respectively. Bars are grouped by method (BW-1 through BW-5, G-e^2, GG-HM, GG-e^2) and colour-coded by contrast condition: pre-contrast (blue), peak-contrast (green), and post-contrast (red).

Table 1. Summary of slope (m), coefficient of determination (R^2), and coefficient of variation (CoV) for all diameter measurement methods across pre-contrast, peak contrast, and post-contrast conditions. Methods are grouped as binary mask based and model-based. Bold values indicate slope agreement within 10% of unity ($0.9 \leq m \leq 1.1$), strong linearity ($R^2 \geq 0.90$), or high precision (CoV ≤ 0.10). (confirm values with MATLAB.)

	BW-1		**BW-2**		**BW-3**		**BW-4**		**BW-5**		$w_{1/e^2,\mathrm{G}}$		FWHM$_{\mathrm{GG}}$		$w_{1/e^2,\mathrm{GG}}$	
	MIP	MeIP	MIP	MeIP	MIP	MeIP	MIP	MeIP	MIP	MeIP	MIP	MeIP	MIP	MeIP	MIP	MeIP
Slope (m)																
Pre	0.61	0.83	0.48	0.74	0.38	0.66	0.30	0.59	0.24	0.53	0.70	0.73	0.42	0.43	0.64	0.71
Peak	**0.96**	1.14	0.86	**1.06**	0.78	**0.99**	0.71	**0.93**	0.65	0.88	1.15	**1.04**	0.66	0.61	**0.98**	**0.98**
Post	**0.92**	1.11	0.82	**1.03**	0.73	**0.97**	0.63	**0.91**	0.54	0.86	1.12	**1.04**	0.66	0.61	**1.02**	**1.00**
Coefficient of determination (R^2)																
Pre	0.58	0.59	0.54	0.62	0.45	0.64	0.34	0.65	0.24	0.65	0.83	0.84	0.79	0.83	0.87	0.88
Peak	0.88	0.78	0.89	0.82	0.86	0.86	0.81	0.88	0.74	0.89	0.87	**0.95**	**0.95**	**0.95**	**0.98**	**0.97**
Post	0.76	0.77	0.69	0.79	0.57	0.80	0.45	0.80	0.34	0.78	0.89	**0.97**	**0.95**	**0.98**	**0.93**	**0.97**
Coefficient of variation (CoV) (values extracted from bar plots, confirm with MATLAB)																
Pre	0.20	0.20	0.24	0.20	0.29	0.19	0.37	0.19	0.47	0.20	0.26	0.14	0.26	0.15	0.25	0.14
Peak	0.17	0.16	0.20	0.15	0.23	0.17	0.27	0.18	0.32	0.18	0.22	0.16	0.23	0.17	0.21	0.15
Post	**0.10**	0.12	**0.10**	0.12	0.11	0.11	0.15	**0.10**	0.19	**0.09**	0.19	**0.10**	0.19	0.11	0.16	**0.09**

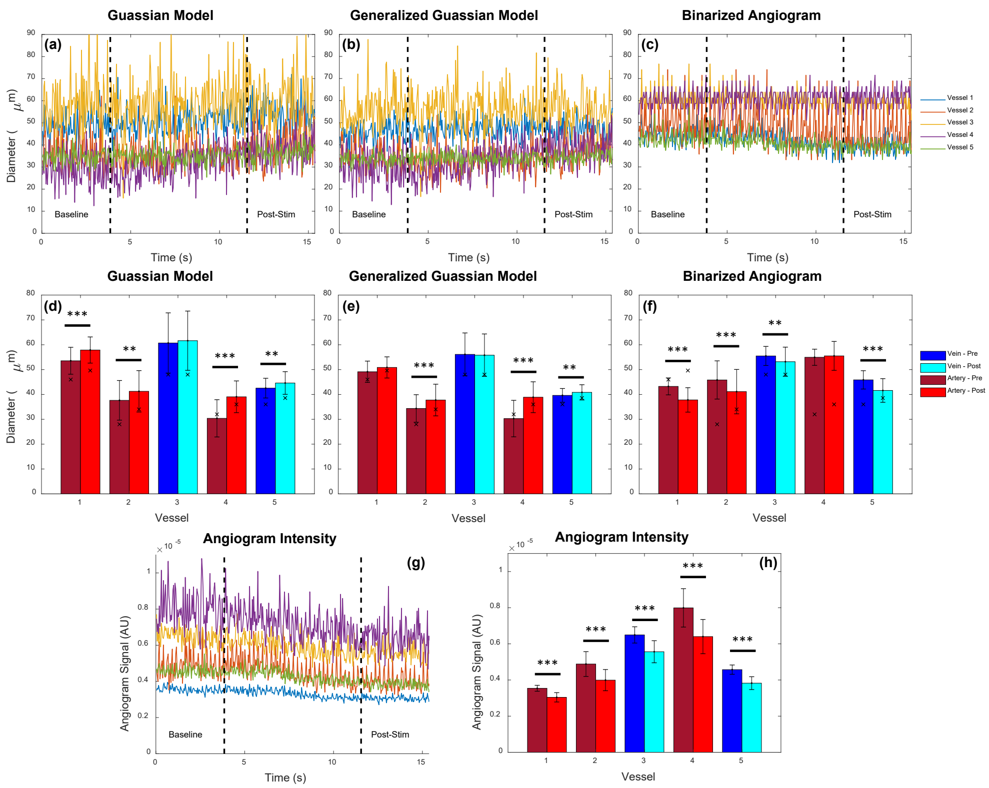

Fig. 8. Functional flicker stimulation results. (a–c) Vessel diameter time series pre- and post-stimulation for (a) Gaussian model, (b) Generalised Gaussian model, and (c) binarised angiogram, across five vessels. Vertical dashed lines indicate stimulation onset and offset. (d–f) Mean diameter per vessel before (dark bars) and after (light bars) stimulation for each method. Veins are shown in blue/cyan and arteries in dark/light red. Significance levels: $^{**}p < 0.01$, $^{***}p < 0.001$. (g) Angiogram intensity time series and (h) mean angiogram intensity per vessel pre- and post-stimulation.

Table 2. Flicker stimulation results. Δd: diameter change (%); ρ: Correlation between diameter and angiogram intensity; ΔI: intensity change (%). Statistically significant values are bold ($^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$).

	Gaussian		**GG**		**Binarised**		**Intensity**
	Δd (%)	ρ	Δd (%)	ρ	Δd (%)	ρ	ΔI (%)
Vessel 1	**+9.2*****	−0.07	+1.5	0.03	**−12.5*****	**0.59*****	**−14.1*****
Vessel 2	**+9.6****	**0.23***	**+9.9*****	0.07	**−10.2*****	**0.80*****	**−18.2*****
Vessel 3	+1.5	**−0.28****	−0.6	**−0.28****	**−4.2****	**0.27****	**−14.3*****
Vessel 4	**+28.5*****	−0.15	**+28.4*****	−0.14	+1.0	**0.46*****	**−19.9*****
Vessel 5	**+4.9*****	**−0.24***	**+3.1****	−0.18	**−9.3*****	**0.52*****	**−16.3*****
Overall ρ		−0.101		−0.098		0.523	

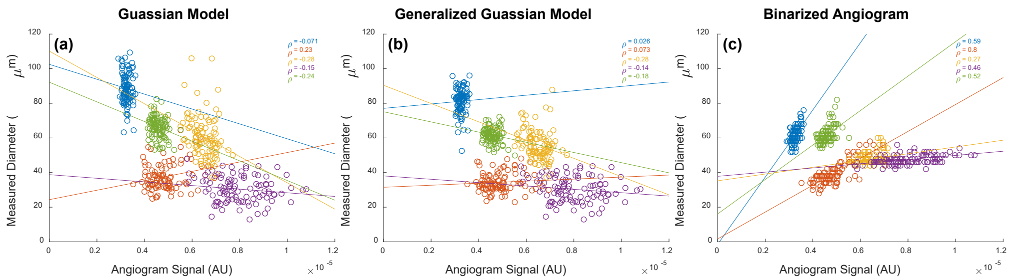

Fig. 9. Measured vessel diameter as a function of angiogram signal intensity during functional flicker stimulation for (a) Gaussian model, (b) generalised Gaussian model, and (c) binarised angiogram. Each colour represents one of five vessels; ρ denotes the Pearson correlation coefficient. The binarised method shows strong positive correlations between diameter and signal intensity, indicating confounding, whereas model-based methods show weak or no correlation.

4. Discussion

4.1. Maximum Intensity or Mean Intensity?

4.2. Future Directions

A future direction of this work to to move toward fully automated segmentation and calculation vessel diameters from enface angiogram maps. While ROIs were manually selected in this work, it is also possible to automatically segment angiographic data using traditional skeletonization methods, which are commonly used for automated OCTA analysis. From a skeletonized image, vessel angle can be computed for further extraction of vascular cross-sections perpendicular to the vessel axis. These cross-sections could then be processed using the model-based methods described here to enable fully automated analysis of 3D OCTA data sets, rather than the semi-automated approach shown here for 2D OCTA data. This type of model-based approach is particularly important for comparing 3D volumes acquired at different time points in the mouse eye as there are often intensity losses caused by drying of the eye, cataract formation, etc. As we have shown in this work, even low levels of intensity loss occurring over a short time window can significantly impact the measured vessel diameter when using a binarization approach, but not when using the models presented here.

4.3. Computation Time

It is clear from the results that model-based approaches outperform traditional diameter measurements based on a binarized image. Nevertheless, the improved linearity and repeatability come at the cost of computation time. Fitting a model to the raw aniogram data is significantly more computationally expensive and may not always be the most appropriate choice if results of the processing are time sensitive. Nevertheless, these approaches are highly parallelizable given the independent nature of the measurements, so it is expected that compuation time could be significantly accelerated with appropriate hardware.

4.4. Threshholding

The results presented here used a basic, global threshold for the binarization of the angiogram data. There are numerous other approaches for binarizing the angiogram, which will most likely each behave differently. For example, angiogram signals can be enhanced with vesselness filters and dynamic thresholds can be used to binarize images with varying SNR. When applying a vesselness filter, it is possible to enhance the shape of the angiogram and omit noise, but it is also possible for such enhancement to overcompensate the signal, leading to larger than expected vessel appearance. Dynamic thresholding can also yield a more uniform angiogram appearance, but similarly risks over- or undercompensating vessels given that threshold level directly affects the measured diameter. Further analysis of these other methods would be required to determine their efficacy in returning accurate and precise diameter values.

4.5. Literature Comparison

Discuss the paper [12]: 'Maximum value projection produces better enface OCT angiograms than mean value projection'.

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